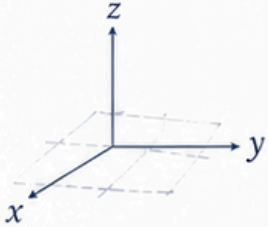
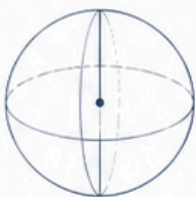
 π

3.1415926535...



$$V = \frac{1}{3} \sqrt{2} a^3$$



$$S = 4\pi r^2$$

Rashtriya Bal Vigyan Pradarshani (RBVP)

2026 – 27

THEME

RECREATIONAL MATHEMATICAL MODELLING

PROJECT TITLE

Reuleaux Tetrahedron



Team Leader : Avadh Prajapati



Co-Leader : Ridham Varma



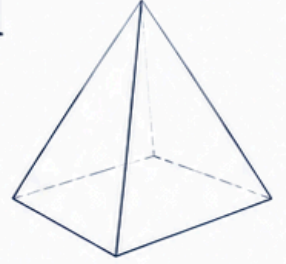
School : PM SHRI Jawahar Navodaya
Vidyalaya, Mehsana



Year : 2026 – 27



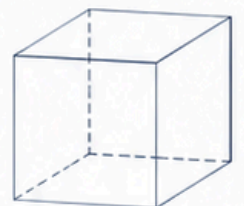
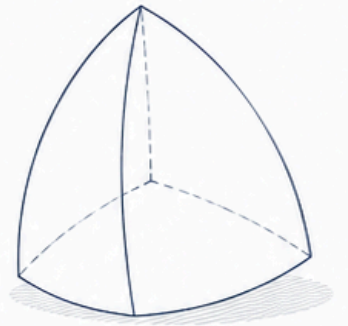
Venue : Gujarat



Euler's Formula

$$V - E + F = 2$$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$



$$a^2 + b^2 = c^2$$

"Mathematics is the language of the universe."

– Galileo Galilei

Certificate

This is to certify that this write-up is prepared for the presentation of the project 'Reuleaux Tetrahedron' at RBVP by Avadh Prajapati.

Acknowledgement

I sincerely thank my teachers, parents and mentors for their guidance and support in completing this project and exploring advanced geometry through 3D printing.

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Introduction

The Reuleaux Tetrahedron is a convex three-dimensional body derived from a regular tetrahedron by replacing each face with a spherical patch centered at the opposite vertex. It demonstrates symmetry, convexity and spatial reasoning.

Objectives

- Study advanced geometry.
- Understand curved solids.
- Demonstrate mathematical modelling using CAD and 3D printing.
- Encourage innovation through mathematics.

Mathematical Derivations

Regular tetrahedron: Vertices = 4, Edges = 6, Faces = 4. Let edge length = a . Each curved face belongs to a sphere of radius $r=a$ centred at the opposite vertex. The model is formed by intersecting these spherical regions. This illustrates convex geometry, spherical geometry and symmetry.

Geometric Properties

Property | Description

Vertices: 4

Edges (underlying tetrahedron): 6

Curved Faces: 4 spherical patches

Symmetry: Tetrahedral

Geometry: Convex, 3D

Verified Applications

Education: teaching advanced geometry and spatial visualization.

Research: convex/computational geometry.

CAD & 3D Printing: modelling and rapid prototyping.

Computer Graphics: geometric algorithms and mesh testing.

Design Inspiration: architecture and engineering concept exploration. **Future**

Scope

Potential inspiration for optimized structural systems, sustainable architecture, AI-assisted geometric optimization, advanced additive manufacturing and space-structure research. These represent future possibilities rather than established commercial uses.

Advantages & Limitations

Advantages: Improves visualization, links mathematics with technology, promotes creativity.

Limitations: Primarily educational and research-oriented; direct industrial use of the exact shape is limited.

References

1. H.S.M. Coxeter, Introduction to Geometry.
2. George Hart, mathematical sculpture resources.
3. Computational Geometry: Algorithms and Applications (de Berg et al.).
4. NCERT Mathematics resources.
5. CAD/3D printing educational documentation.